FINAL EXPLANATION: MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: MATHEMATICS GRADE 10 – SUB-STRAND 1.1: REAL NUMBERS	
Student Assessment Document	
Student Name	______________________________________
Class	______________________________________
Date	______________________________________

INSTRUCTIONS FOR STUDENTS

FINAL EXPLANATION — REAL NUMBERS UNIT
Grade 10 Mathematics | Lesson 7 Final Synthesis

You have now completed all 7 lessons on Real Numbers. This is your chance to bring everything together and show what you understand.

YOUR TASK: Write a complete, well-reasoned explanation that answers the unit's driving question:
"Why can some numbers be written as exact fractions and others cannot — and how does knowing a number's 'type' help us solve everyday problems in Kenya?"

Use this document as your guide. Work through each part in order. For every part, refer back to your Summary Table, your class notes, and the evidence you collected during lessons. Use specific examples, mathematical notation, and real Kenyan contexts wherever you can.

REMEMBER THE PIZZA PHENOMENON: Throughout your explanation, connect your ideas back to the Unequal Pizza Slices problem you explored at the start of the unit. Think about why some pizza slices could be described with neat fractions and others could not — and what that tells us about the real number system.

PART 1: Classification of Whole Numbers — Odd, Even, Prime, and Composite

Explain what you have learned about classifying whole numbers. Your response must include ALL of the following: – **ODD and EVEN numbers: Give the definition, describe the digit pattern, and explain the divisibility**	After completing Lesson 1, I can classify any whole number as odd, even, prime, or composite — and I know why these categories matter in everyday life. ODD AND EVEN NUMBERS Even numbers are whole numbers that can be divided exactly by 2 with no remainder. They always end in 0, 2, 4, 6, or 8. For example: 2, 4, 6, 8, 10, 14, 20. If a Nairobi teacher wants to pair all 30 students in a class into groups of 2, she can do so perfectly because 30 is even (30 ÷ 2 = 15 pairs, no remainder).

rule. Provide at least one Kenyan real-life example of why even or odd numbers matter (e.g., pairing students for group work, arranging chairs in a Nairobi classroom, sharing mandazi equally). **– PRIME and COMPOSITE numbers: Give the definition of each, list the first 10 prime numbers, and explain why 1 is a special case. Give at least one real-life use of prime numbers (e.g., secure M-Pesa transactions rely on prime-number cryptography).** **– EVIDENCE: Refer to what you discovered in Lesson 1 when you sorted and classified numbers using hands-on activities.** **Key vocabulary to use: factor, divisible, remainder, prime factorisation.**	Odd numbers cannot be divided exactly by 2 — there is always a remainder of 1. They end in 1, 3, 5, 7, or 9. For example: 1, 3, 5, 7, 9, 11, 15. If 15 mandazi are shared between 2 friends, one friend gets an extra piece (15 ÷ 2 = 7 remainder 1), so the sharing is not perfectly equal. Real-life connection: Knowing whether the number of items is odd or even tells us immediately whether equal pairing is possible — before we even start dividing. PRIME AND COMPOSITE NUMBERS A prime number has exactly two factors: 1 and itself. It cannot be divided evenly by any other whole number. The first ten prime numbers are: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29. Notice that 2 is the only even prime — every other even number has 2 as an additional factor, making it composite. A composite number has more than two factors. For example, 12 is composite because its factors are 1, 2, 3, 4, 6, and 12. Any composite number can be broken down into a product of prime factors (prime factorisation): 12 = 2 × 2 × 3. Special case — the number 1: The number 1 is neither prime nor composite. It has only one factor (itself), so it does not meet the definition of prime (which requires exactly two factors). Real-life use: Prime numbers protect our digital information. When you use M-Pesa or buy data bundles on your phone, the security system uses very large prime numbers to encrypt your personal details — making it almost impossible for anyone to decode your information without the correct key. Evidence from Lesson 1: When we sorted number cards into groups during the classification activity, I discovered that composite numbers always appeared more frequently than prime numbers as numbers get larger — and that every composite number could always be rebuilt from its prime "building blocks."

PART 2: Rational Numbers — Fractions, Decimals, and the Numbers We Can Name Exactly	
Explain what rational numbers are and how you recognise them. Your response must include ALL of the following: **– DEFINITION: State the formal definition of a rational number using fraction notation (p/q where q ≠ 0). Explain in your own words what the word 'rational' means and where**	DEFINITION A rational number is any number that can be written as a fraction p/q, where p and q are both integers and q is not equal to zero. The word 'rational' comes from the word 'ratio' — a comparison of two quantities. So a rational number is simply a number that can be expressed as a ratio of two integers. IMPORTANT: The denominator can never be zero. Division by zero is undefined in mathematics. TYPES OF RATIONAL NUMBERS — with examples: – Whole numbers: Every whole number is rational because it can be written over 1. Example: 8 = 8/1. – Integers (including negatives): −5 = −5/1, so negative whole numbers are also rational. – Proper fractions: ½, ¾, 2/5 — the numerator is smaller than the denominator.

it comes from. – **TYPES: Show that rational numbers include whole numbers, integers, proper fractions, improper fractions, terminating decimals, and repeating decimals. Give at least one example of each type.** – **DECIMAL PATTERNS: Explain the difference between a terminating decimal and a repeating decimal. Use your calculator evidence from Lesson 2 to support your explanation.** – **KENYAN REAL-LIFE EXAMPLES: Use at least two examples from Kenyan daily life (e.g., prices in shillings, recipe measurements for ugali or chapati, market weight measurements in Gikomba Market, percentage marks on a school report).** – **PIZZA CONNECTION: Explain which pizza slices from the anchoring phenomenon were rational, and why.** **Key vocabulary to use: ratio, terminating, repeating, numerator, denominator, integer.**	– Improper fractions: 7/3, 11/4 — the numerator is larger than the denominator. – Terminating decimals: 0.5 = ½; 0.25 = ¼; 0.125 = ⅛. These decimals end after a fixed number of digits. – Repeating decimals: 0.333… = ⅓; 0.666… = ⅔; 0.142857142857… = 1/7. These decimals go on forever but follow a repeating pattern. DECIMAL PATTERNS Terminating decimals stop. When I typed ½ into a calculator in Lesson 2, I got 0.5 — it stopped there. When I typed ¼ I got 0.25. These come from fractions whose denominators have only 2 and 5 as prime factors. Repeating decimals continue forever but with a pattern. When I typed ⅓, I got 0.333333… — the digit 3 repeats endlessly. When I typed 1/7, I got 0.142857142857… — a block of six digits that repeats. Both of these are still rational because there IS a fraction that equals them exactly. KENYAN REAL-LIFE EXAMPLES 1. Prices in shillings: A packet of unga (maize flour) costs Ksh 145.50. This is the rational number 145½, or 291/2 in fraction form. It can be written exactly as a fraction. 2. Recipe measurements: A recipe for chapati uses ¾ cup of water. This fraction is rational — it tells us exactly how much water to add. A cook in Kericho can measure this precisely. 3. School marks: If you score 56 out of 80, your fraction is 56/80 = 7/10 = 70%. This is rational — it terminates at 0.7. 4. Market weights: A trader at Gikomba Market sells 2.5 kg of dried beans. 2.5 = 5/2, a rational number. PIZZA CONNECTION In the Unequal Pizza Slices problem, the slices that were described using fractions like ¼ or ⅛ are rational numbers. For example, if a pizza is divided into 4 equal slices, each slice is ¼ of the whole. When I convert ¼ to a decimal, I get 0.25 — a terminating decimal. Similarly, ⅓ of a pizza = 0.333… a repeating decimal, but it is still rational because it equals the fraction ⅓. Both of these slices can be described exactly using the fraction form p/q.

PART 3: Irrational Numbers — The Decimals That Never End and Never Repeat

Explain what irrational numbers are and how you recognise them. Your response must include ALL of the following: – **DEFINITION: State the formal definition of an irrational number. Explain clearly why the word 'irrational' does NOT mean 'crazy' — it means 'cannot be expressed as a ratio.'** – **KEY EXAMPLES: Discuss at least three irrational numbers: √2, √3, and π (pi). For each one, state its approximate decimal value, explain why it is irrational, and describe where it appears in real life.** – **CALCULATOR EVIDENCE: Describe what you observed when you typed √2 and π into a calculator during your lessons. What did the decimal do? How is that different from ⅓?** – **KENYAN REAL-LIFE EXAMPLES: Use at least two Kenyan contexts where irrational numbers appear (e.g., the radius and circumference of a circular maize storage silo in Eldoret; the diagonal of a square football pitch in Kasarani Stadium; the arc length of a pizza slice).** – **PIZZA CONNECTION: Explain which part of the pizza phenomenon**	DEFINITION An irrational number is a number that CANNOT be written as a fraction p/q where p and q are integers and $q \neq 0$. The prefix 'ir-' means 'not', so irrational means 'not a ratio.' It does not mean the number is strange or impossible — irrational numbers exist on the number line just like rational numbers do. In fact, we can see irrational numbers in physical measurements every day. The key feature of an irrational number's decimal form is that it: ✗ Never terminates (never stops) ✗ Never repeats (no block of digits ever repeats in a pattern) KEY EXAMPLES OF IRRATIONAL NUMBERS 1. $\sqrt{2} \approx 1.41421356237\ldots$ The square root of 2 is the length of the diagonal of a square with sides of 1 unit. When I typed √2 into a calculator, I got a decimal that kept going with no repeating pattern. No matter how many decimal places I calculated, it never terminated and never repeated. This means √2 cannot equal any fraction p/q. Real-life use: If the side of a square courtyard at a school in Nairobi is 1 metre, the diagonal across it is exactly √2 metres — approximately 1.414 metres. You cannot measure this perfectly with a fraction ruler. 2. $\sqrt{3} \approx 1.73205080757\ldots$ Similarly, √3 never terminates and never repeats. It appears in calculations involving equilateral triangles — for example, the height of a triangular tent frame used by Maasai herdsmen in the Rift Valley. 3. π (pi) $\approx 3.14159265358979\ldots$ Pi is the ratio of a circle's circumference to its diameter. Although it looks like a ratio, both the circumference and the diameter cannot both be rational at the same time — so π itself is irrational. Mathematicians have calculated π to trillions of decimal places and it never terminates or repeats. Real-life use: To find the circumference of a circular maize storage silo in Eldoret with a diameter of 3 metres, we calculate $C = \pi \times 3 \approx 9.4248\ldots$ metres. This is an approximation — we can never write the exact circumference as a terminating decimal or simple fraction. CALCULATOR EVIDENCE In Lesson 3, I typed √2 into a calculator and saw: 1.41421356237… I also typed ⅓ and saw: 0.333333333… Here is the critical difference: – ⅓ = 0.333… is non-terminating BUT it repeats (the digit 3 repeats forever in a pattern). This makes it RATIONAL. – √2 = 1.41421356… is non-terminating AND non-repeating (no pattern ever appears). This makes it IRRATIONAL. KENYAN REAL-LIFE EXAMPLES 1. Kasarani Stadium football pitch: If the pitch is a rectangle 100 m × 64 m, the diagonal $= \sqrt{(100^2 + 64^2)} = \sqrt{(10000 + 4096)} = \sqrt{14096} \approx 118.72\ldots$ m. This is irrational — it cannot be written as an exact fraction.

involved an irrational number, and why that slice cannot be described with a neat fraction. **Key vocabulary to use: square root, non-terminating, non-repeating, circumference, diagonal, approximate.**	2. Circular water tanks in Kisumu: Many households use cylindrical water tanks. To calculate how much pipe goes around the top rim of a tank with diameter 1.2 m: Circumference = $\pi \times 1.2 \approx 3.7699\ldots$ m. Because π is irrational, this measurement is always approximate. PIZZA CONNECTION In the Unequal Pizza Slices problem, the slice described using $\sqrt{2} \div 2 \approx 0.7071\ldots$ is an example of an irrational number. Even though the slice exists physically — you can hold it, eat it, and see it — you cannot describe its size with a neat fraction. The decimal 0.70710678… never ends and never repeats. This is what makes irrational numbers surprising: they are real, measurable quantities, but they escape the neat fraction system we use for rational numbers.

PART 4: Reciprocals — Flipping Numbers and Why It Matters	
Explain the concept of reciprocals and their properties. Your response must include ALL of the following: – **DEFINITION: Define the reciprocal of a number. Explain the rule for finding the reciprocal of a fraction, a whole number, and a decimal.** – **KEY PROPERTY: State and explain the fundamental property of reciprocals: a number multiplied by its reciprocal always equals 1. Show this with at least three worked examples using numbers (including at least one fraction and one irrational number approximation).** – **RATIONAL vs. IRRATIONAL RECIPROCALS: Explain what happens to the reciprocal of a rational number (it stays rational) and the reciprocal of an irrational number (it stays**	DEFINITION OF RECIPROCAL The reciprocal of a number is what you multiply it by to get 1. It is also called the multiplicative inverse. To find the reciprocal, you flip the numerator and the denominator of a fraction. Rules: – Reciprocal of a fraction p/q → flip it to get q/p. Example: reciprocal of ¾ = 4/3. – Reciprocal of a whole number n → write it as n/1, then flip: reciprocal of 5 = 1/5. – Reciprocal of a decimal → convert to a fraction first, then flip. Example: 0.25 = ¼, so reciprocal = 4/1 = 4. – Special case: The number 0 has NO reciprocal because 1/0 is undefined. KEY PROPERTY: A NUMBER × ITS RECIPROCAL = 1 Worked examples: 1. ¾ × 4/3 = 12/12 = 1 ✓ 2. 5 × 1/5 = 5/5 = 1 ✓ 3. $\sqrt{2} \times (1/\sqrt{2}) = \sqrt{2}/\sqrt{2} = 1$ ✓ (even irrational numbers have this property!) This property is true for ALL non-zero numbers — rational and irrational alike. RATIONAL vs. IRRATIONAL RECIPROCALS – The reciprocal of a rational number is always rational. Example: reciprocal of ⅔ = 3/2 = 1.5. This is still a fraction (rational). – The reciprocal of an irrational number is always irrational. Example: reciprocal of $\sqrt{2} = 1/\sqrt{2} \approx 0.7071\ldots$ This decimal never terminates and never repeats, so $1/\sqrt{2}$ is irrational. This is an important pattern: changing a number into its reciprocal does not change whether it is rational or irrational. PIZZA AND REAL-LIFE CONNECTION Reciprocals answer the question: "How many of this piece fit into the whole?"

irrational). Give examples. – **PIZZA AND REAL-LIFE CONNECTION: Connect reciprocals back to the pizza phenomenon. If you know one slice is ⅛ of the pizza, how many slices fill the whole? Show how this uses the reciprocal. Use at least one additional Kenyan example (e.g., if a jembe (hoe) covers ⅓ of an acre per hour, how many hours to cover the whole acre?).** – **DIVISION RULE: Explain why dividing by a number is the same as multiplying by its reciprocal. Show one worked example.** **Key vocabulary to use: reciprocal, multiplicative inverse, product, undefined.**	Pizza example: If each slice is ⅛ of the pizza, how many slices make a whole pizza? Answer: 1 ÷ ⅛ = 1 × 8/1 = 8 slices. The reciprocal of ⅛ is 8. This is exactly the question posed in the pizza phenomenon — and the answer comes directly from the reciprocal! Now notice: if a slice is √2 ÷ 2 of the pizza (an irrational amount), how many such slices fill the whole? Answer: 1 ÷ (√2/2) = 2/√2 ≈ 1.4142… slices. This is an irrational number — you can never have a whole-number count of perfectly equal irrational-sized slices that fill the pizza exactly. This explains the puzzle at the heart of our phenomenon. Kenyan farming example: A farmer in Nakuru uses a jembe to dig ⅓ of an acre per hour. How many hours does she need to dig a full acre? Answer: 1 ÷ ⅓ = 1 × 3/1 = 3 hours. The reciprocal of ⅓ is 3. This is a practical use of reciprocals in agricultural planning. DIVISION RULE Dividing by a number is the same as multiplying by its reciprocal. This is because division and multiplication by the reciprocal are inverse operations. Example: How many 2/5 kg portions of githeri can you get from 4 kg? 4 ÷ 2/5 = 4 × 5/2 = 20/2 = 10 portions. So 10 people can each receive a 2/5 kg serving — found by multiplying by the reciprocal 5/2.

PART 5: The Real Number System — Putting It All Together and Answering the Driving Question	
Bring all your learning together to answer the unit's driving question fully and precisely. Your response must include ALL of the following: – **THE REAL NUMBER SYSTEM: Draw or describe the nested structure of real numbers. Show how natural numbers ⊂ whole numbers ⊂ integers ⊂ rational numbers, and how rational numbers and irrational numbers together form the real numbers. Use**	THE REAL NUMBER SYSTEM — NESTED STRUCTURE The real number system is organised like a set of nesting containers, where each inner set is completely contained in the next: Natural Numbers (N): 1, 2, 3, 4, 5… (counting numbers) ⊂ Whole Numbers (W): 0, 1, 2, 3, 4… (natural numbers + zero) ⊂ Integers (Z): …−3, −2, −1, 0, 1, 2, 3… (whole numbers + negatives) ⊂ Rational Numbers (Q): all numbers expressible as p/q (integers + fractions + terminating and repeating decimals) Irrational Numbers (Q'): numbers that CANNOT be expressed as p/q (e.g., √2, √3, π) Rational Numbers (Q) + Irrational Numbers (Q') = Real Numbers (R) The symbol ⊂ means 'is a subset of' — every element of the inner set also belongs to the outer set. For example, every integer is also a rational number (because any integer n = n/1), but not every rational number is an integer.

the correct mathematical subset symbol (⊂) and explain what it means. – **ANSWER THE DRIVING QUESTION: In at least one clear paragraph, answer: 'Why can some numbers be written as exact fractions and others cannot — and how does knowing a number's type help us solve everyday problems in Kenya?' Use evidence from all 7 lessons.** – **CLASSIFY A MIXED LIST: Classify each of the following numbers and justify your answer: (a) 0.75, (b) √5, (c) −4, (d) π, (e) 22/7, (f) 0.101001000100001…** – **PIZZA PHENOMENON RESOLUTION: Write a final paragraph that fully resolves the pizza phenomenon. Explain why some cuts gave rational slice sizes and others gave irrational ones, what reciprocals revealed about fitting slices into the whole, and what this tells us about the real number system.** – **REFLECTION: Write 2–3 sentences about one thing in this unit that surprised you, and how your thinking about numbers has changed.** **Key vocabulary to use: subset,**	ANSWERING THE DRIVING QUESTION Some numbers can be written as exact fractions because their decimal forms either terminate or repeat in a fixed pattern — and any such decimal can be converted back into a fraction p/q. These are rational numbers. Other numbers, like √2 and π, have decimal forms that go on forever with absolutely no repeating pattern. No fraction can ever equal them exactly — they are irrational. Knowing a number's type is not just a mathematical exercise; it determines what we can measure exactly and what we must approximate. In everyday Kenyan life, when a mama mboga at Gikomba Market sells goods in fractional kilograms, she works with rational numbers and can calculate change exactly. When an engineer in Nairobi calculates the circumference of a circular water reservoir, she uses π and knows her answer is an approximation — she must decide how many decimal places are precise enough for safety. Understanding the real number system helps us know when to expect exact answers and when to work with carefully chosen approximations. CLASSIFY THE NUMBERS (a) 0.75 → RATIONAL. 0.75 = ¾. It is a terminating decimal expressible as a fraction. (b) √5 ≈ 2.2360679… → IRRATIONAL. Its decimal never terminates and never repeats. It cannot be written as p/q. (c) −4 → RATIONAL (and an integer). −4 = −4/1. All integers are rational. (d) π ≈ 3.14159265… → IRRATIONAL. Pi's decimal never ends and never repeats. Although 22/7 is sometimes used as an approximation for π, they are not equal. (e) 22/7 → RATIONAL. It is already in the form p/q (22 and 7 are both integers, 7 ≠ 0). Note: 22/7 ≈ 3.142857142857… — a repeating decimal, confirming it is rational. It is NOT equal to π. (f) 0.101001000100001… → IRRATIONAL. The decimal never terminates and there is no fixed repeating block (the pattern of zeros grows by one each time). It cannot be written as p/q. PIZZA PHENOMENON — FINAL RESOLUTION At the start of this unit, we were puzzled by the Unequal Pizza Slices problem: two slices of pizza that both physically exist, yet one could be described with a neat fraction and the other could not. We now have a complete explanation. When a pizza is divided into 4 equal slices, each slice is ¼ of the whole — a rational number whose decimal (0.25) terminates cleanly. When we tried to express a slice in terms of √2 ÷ 2, we got a decimal (≈ 0.7071…) that never ends and never repeats — an irrational number. The physical slice exists, but it escapes the rational number system. Reciprocals deepened our understanding: for the ¼ slice, the reciprocal is 4 — exactly 4 slices fill the whole pizza. For the irrational slice (√2/2), the reciprocal is √2 ≈ 1.4142… — which tells us that 1.4142… such slices would fill the pizza. You can never have a whole-number count of perfectly equal irrational slices that tile the pizza exactly. This is the deeper truth the phenomenon revealed: the real number system contains both types of numbers because measurement itself — of length, area, circumference, and more — produces both. The rational numbers are not enough on their own; irrational numbers arise naturally and necessarily whenever we measure with geometric precision. MY REFLECTION One thing that surprised me in this unit was discovering that a number like √2 genuinely cannot be written as any fraction — no matter how many digits I use in the numerator and denominator, I can only ever get an approximation. Before this unit, I thought all decimals could eventually be turned into fractions. My thinking has changed: I now understand that the number line is actually filled with far more irrational numbers than rational ones, even though the rational numbers are the ones we use most in daily calculations. I will never look at a circle — or a pizza — the same way again.

real number system, classify, rational, irrational, reciprocal, terminating, non-repeating, approximate.	

RUBRIC			
Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**
Classification of Real Numbers (Odd/Even, Prime/Composite, Rational/Irrational)	Accurately defines and distinguishes all number types using correct mathematical language. Correctly classifies all items in the mixed list with clear justification. Explains the nested subset structure ($N \subset W \subset Z \subset Q$) using the $\subset$ symbol correctly. Identifies the special cases (1 is neither prime nor composite; $22/7 \neq \pi$).	Defines most number types correctly with minor language imprecision. Classifies most items in the mixed list correctly, with partial justification. Describes the subset structure in words but may omit the formal $\subset$ notation. Identifies at least one special case.	Definitions are incomplete or confused (e.g., mixes up irrational and non-terminating). Classifies fewer than half the items correctly or provides no justification. Does not describe the subset structure. Special cases are missed or incorrectly stated.
Understanding of Rational and Irrational Numbers (Decimal Patterns and Evidence)	Clearly explains the difference between terminating, repeating, and non-terminating non-repeating decimals with correct examples for each type. Uses calculator evidence from lessons to support claims. Correctly explains why 22/7 is rational and π is irrational even though their decimal approximations look similar.	Explains the difference between rational and irrational decimals with mostly correct examples. References lesson evidence in a general way. May confuse 22/7 and π or not fully explain the distinction.	Struggles to distinguish decimal types or uses examples incorrectly (e.g., claims 0.333… is irrational). Little or no reference to lesson evidence. Does not address 22/7 vs. π.
Reciprocals — Definition, Properties, and Application	Correctly defines the reciprocal and states the multiplicative inverse property ($a \times 1/a = 1$) with at least three worked examples including a fraction and an irrational number. Explains why 0 has no reciprocal. Correctly explains the division rule (dividing = multiplying by the reciprocal) with a worked example. Connects reciprocals clearly to the pizza phenomenon.	Defines reciprocal correctly and demonstrates the property with two or more examples. May omit the case of 0 or the irrational reciprocal example. Explains the division rule but may lack a fully worked example. Makes a connection to the pizza phenomenon.	Definition of reciprocal is partially correct or incomplete. Fewer than two correct examples of the property. Division rule is absent or incorrectly stated. No clear connection to the pizza phenomenon.
Kenyan Real-Life Applications and Contextual Reasoning	Integrates at least four distinct Kenyan real-life examples across different parts of the explanation (e.g., M-Pesa security, maize flour prices, Kasarani Stadium diagonal, circular water tanks in Kisumu, farming with a jembe). Each example is mathematically accurate and clearly linked to the number concept being	Includes two or three Kenyan real-life examples that are mostly accurate and linked to relevant number concepts. Examples may be repetitive or lack full mathematical explanation.	Includes one or no Kenyan real-life example, or examples used are not clearly connected to the mathematical concept being discussed. May use generic, non-Kenyan contexts instead.

	explained.		
Answering the Driving Question and Resolving the Pizza Phenomenon	Provides a complete, precise, and well-evidenced answer to the driving question, explaining both why some numbers cannot be exact fractions and how number classification supports real-world problem-solving. Fully resolves the pizza phenomenon by connecting rational/irrational slice descriptions and the reciprocal's role in determining how many slices fill the whole. Reflection shows genuine conceptual change.	Answers the driving question clearly but may address only one part in depth (the 'why' or the 'how'). Resolves the pizza phenomenon with some mathematical detail but may not fully connect reciprocals to the resolution. Reflection is present but surface-level.	Answer to the driving question is vague or incomplete — does not explain why irrational numbers cannot be fractions, or does not connect number type to real-world use. Pizza phenomenon resolution is missing or inaccurate. Reflection is absent or unrelated to mathematical learning.